Optimisation

1 For each of the following functions:

- **i** Find any stationary points of the function in the given domain.
- **ii** Find the absolute maximum value of the function in the given domain.
- **iii** Find the absolute minimum value of the function in the given domain.

a $f:[-7,5]\to\mathbb{R}$ given by $f(x)=-6x^2-12x+90$.

b $f:[3,8]\to\mathbb{R}$ given by $f(x)=2x^2-2x-24$.

c $f:[-7,3]\to\mathbb{R}$ given by $f(x)=8x^2+32x-96$.

d $f:[-7,-4]\to\mathbb{R}$ given by $f(x)=-4x^2-8x+60$.

e $f:[4,14]\to\mathbb{R}$ given by $f(x)=x^3-19x^2+104x-137$.

2 Consider the function $f(x)=2x-4\sqrt{x}$.

a Find the stationary point of the function.

b Find the absolute minimum value of the function within the range $\frac{1}{2}\leq x\leq 2$.

3 Consider the parabola with equation $y=x^2-4x+6$.

a Will this parabola have an absolute minimum or absolute maximum value? Explain your answer.

b Find this absolute value of the function.

4 Consider the parabola with equation $y=-2x^2+10x-12$.

a Will this parabola have an absolute minimum or absolute maximum value? Explain your answer.

b Find this absolute value of the function.

Applications

5 The height at time t of a ball thrown upwards is given by the equation $h=59+42t-7t^2$.

a How long does it take the ball to reach its maximum height?

b Find the height of the ball at its highest point.

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6 A parabolic satellite dish is pointing straight up. Along a cross-section that passes through the centre of the dish, the height above ground is given by the following equation:

$$y = \frac{1}{100}\left(x^2 - 100x\right) + 50$$

a Find $\frac{dy}{dx}$.

b For what value of x is $\frac{dy}{dx}$ equal to 0?

c How far above the ground is the dish at its lowest point?

7 A potato farmer finds that the yield per square metre when spacing his plants between $0.5\ \text{m}$ and $3.0\ \text{m}$ approximates the following equation:

$$y = -\frac{x^2}{4} + \frac{7x}{20}$$

a Find $\frac{dy}{dx}$.

b For what value of x is $\frac{dy}{dx}$ equal to 0?

c Find the maximum possible yield. Round your answer to two decimal places.

8 The sum of two whole numbers is 24. Let one of the numbers be x.

a Let y represent the product of the numbers. Form an expression for y in terms of x.

b Find the value of x that will result in the greatest product of the two numbers.

c Find the greatest possible product of the two numbers.

9 A rectangle is constructed such that the sum of its length and width is 20. Let the length of the rectangle be x.

a Form an expression for A, the area of the rectangle, in terms of x.

b What length of rectangle will result in the maximum possible area?

c What is the width of the rectangle with maximum area?

10 A rectangular paved labyrinth is to be built in the local Botanic Gardens. The stonemasons have $336\ \text{m}$ of rare marble for the perimeter, but they want to maximise the area of the labyrinth. Let x be the width of the labyrinth.

a Find an expression for the length of the labyrinth in terms of x.

b Form an expression for A, the area of the labyrinth.

c What width will allow for the greatest possible area?

d What is the greatest possible area of such a labyrinth?

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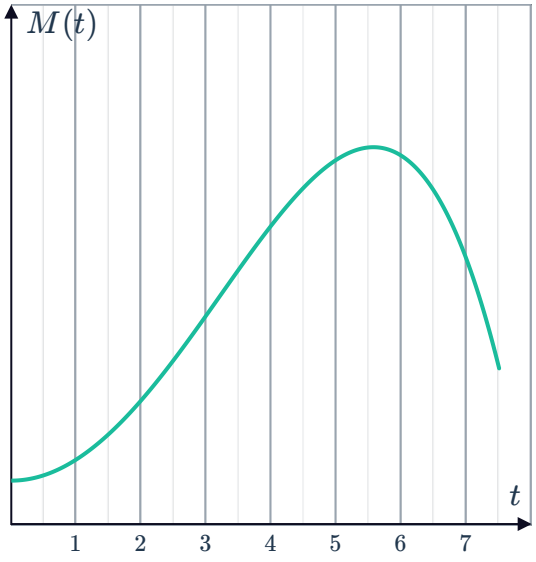
11 A ball is thrown upwards from a height of $4\ \text{m}$ with an initial velocity of $44\ \text{m/s}$. The height at time t of the ball is given by the equation $h = -16t^2 + 44t + 4$.

- **a** How long does it take the ball to reach the ground? Round your answer to two decimal places.
- **b** How long does it take the ball to reach its maximum height? Give your answer to two decimal places.
- **c** Find the height of the ball at its highest point, correct to two decimal places.

12 A pool is being emptied, and the volume of water $V\ \text{L}$ left in the pool after t minutes is given by the equation $V = 1500\,(11 - t)^3$, for $0 \leq t \leq 11$.

- **a** State the rate of change of the volume after t minutes.
- **b** At what rate is the volume of water in the pool changing after 10 minutes?
- **c** State the time t at which the pool is emptying at the fastest rate.

13 Web developers want to check that a site doesn't crash when exam results are released. They look back on the previous year and find that the function $M(t) = 128 + 62t^2 - t^4$ accurately approximated the number of students, M, logged onto the site at any time over the first 7.5 hours, $0 \leq t \leq 7.5$.
The graph of the function over the region $0 \leq t \leq 7.5$ is shown below:



- **a** Initially, how many students logged onto to check their results?
- **b** How many students logged on to view their results at the end of the 7.5 hour time period? Round your answer to the nearest whole number.
- **c** Find the values of t for which $M'(t) = 0$.
- **d** Find the maximum number of students logged onto the website at any one time in the first 7.5 hours. Round your answer to the nearest whole number.
- **e** Find the time t at which the students were logging onto the site most rapidly.

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14 Maximilian is fencing off a rectangular section of his backyard to use for a vegetable garden. He uses the existing back wall and has 28 m of fencing to create the other three sides. He wishes to make the area for vegetables as large as possible.

a Write an equation for the length, L, of the vegetable garden in terms of the width, W.

b Hence write an equation for the area, A, of the vegetable garden in terms of the width, W.

c Find the width, W, that could maximise the area of the vegetable garden.

d Find the length of the vegetable garden that maximises the area.

15 A rectangular box is to be made with the following constraints:

- The length must be 3 times the width
- The total length of the edges must be 192 cm

a Write an equation for the height, h, in terms of the width, w.

b Write an expression for the volume, V, of the box in terms of the width, w, only.

c Find the possible width(s), w, that could maximise the volume of the box.

d Find the maximum volume of the box.

e Hence state the dimensions of the box with the maximum volume.

16 A rectangular sheet of cardboard measuring 72 cm by 45 cm is to be used to make an open box. A square of width w cm is removed from each corner to make the net shown:

a Write an expression for the length of the box in terms of w.

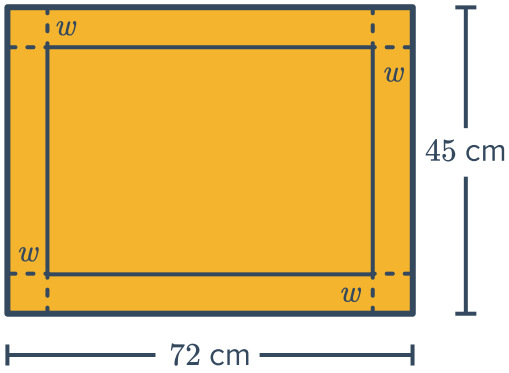
b Write an expression for the width of the box in terms of w.

c Write an expression for the volume, V, of the box in terms of w.

d Find the possible value(s) of w that could maximise the volume of the box.

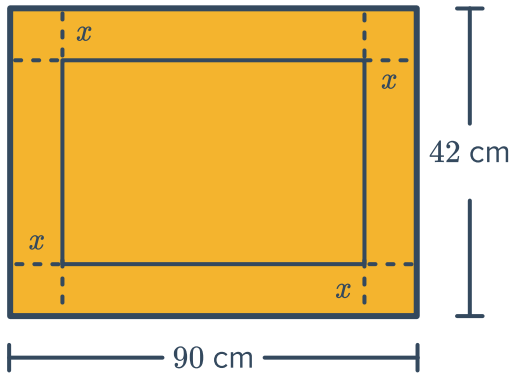
e Determine which value of w maximises the volume of the box.

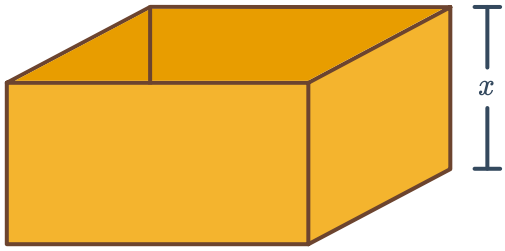
f Hence find the dimensions of the box with the maximum volume.



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17 A box without a lid is to be constructed from a piece of rectangular cardboard that measures 90 cm by 42 cm. Four identical squares will be cut out of the corners of the rectangle to allow the sides to fold up:





Let x be the height of the box, and V be the volume of the box.

a Form an equation for V, in terms of x.

b Find the value of x that maximises the volume of the box.

c Calculate the maximum volume of the box.

18 A rectangular cereal box with a square base and open top is to have a volume of 256 cm^3.

a If the side lengths of the base is x cm, and the height of the box is h cm, express h in terms of x.

b Let the surface area of the open box be represented by S. Find an equation for S in terms of x.

c Find the values of x that will minimise the amount of material required to make the box.

19 A cylindrical tin can with radius r cm and height h cm is to be designed so that the total surface area including the top and bottom is 5π cm^2.

a Express h in terms of r.

b Let V be the volume of the can. Express V in terms of r.

c Find the the radius of the base that will result in the can of the largest volume. Leave your answer in exact form.

20 A manufacturer makes cylindrical tins of volume 600 cm^3. The manufacturer wishes to make tins with the smallest possible surface area that still meet the volume requirement.

a Write an equation for the height, h, of the cylinder in terms of the radius, r.

b Hence write an equation for the surface area, A, of the cylinder in terms of the radius, r.

c Find the radius r that will minimise the surface area.

d Find the minimum surface area.

e Find the height of the cylinder with the minimum surface area.

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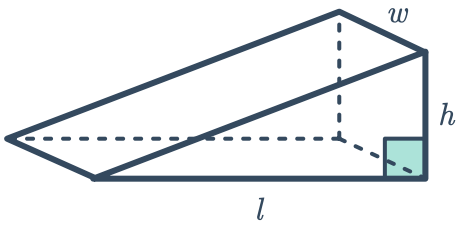
21 A construction company has $2.5\ \text{m}^3$ of concrete to make an access ramp. The ramp is a triangular prism whose triangular faces are right-angled. The height of the ramp is $62.5\ \text{cm}$.

a Find an equation for the length, l, of the ramp in terms of its width, w.

b Find $\frac{dl}{dw}$.

c Are there any values of w for which $\frac{dl}{dw}$ is equal to 0?

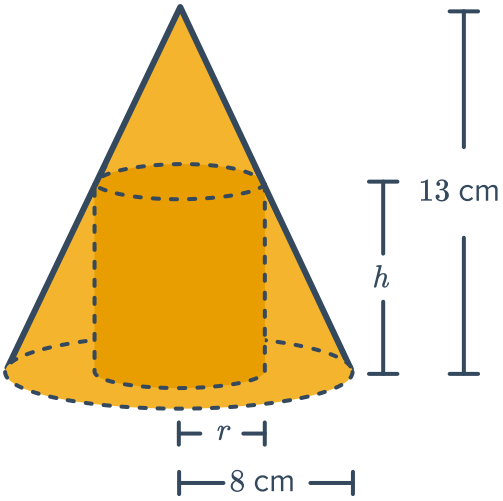
d If the width cannot be less than $80\ \text{cm}$ or greater than $1.6\ \text{m}$, find the minimum length of the ramp.



22 A cylinder of radius r and height h is to be inscribed in a right circular cone with radius $8\ \text{cm}$ and height $13\ \text{cm}$:

a Find an equation that expresses r in terms of h.

b Let V represent the volume of the cylinder. Find the values of h that will result in the cylinder of greatest volume being inscribed in the cone.



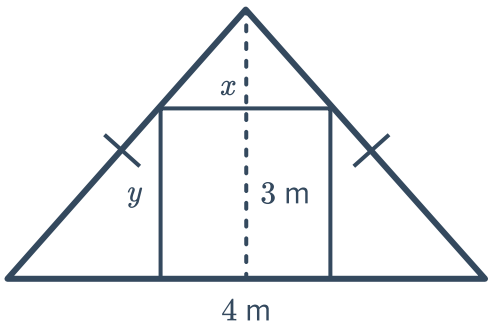
23 A rectangular door that is $x\ \text{m}$ wide and $y\ \text{m}$ tall is to be inscribed into a triangular wall. The wall is isosceles, measuring $4\ \text{m}$ across the base and $3\ \text{m}$ high. The door cannot be shorter that $1.8\ \text{m}$ or narrower than $0.6\ \text{m}$.

a Write an equation for y in terms of x.

b Find an expression for the area, A, of the door in terms of x.

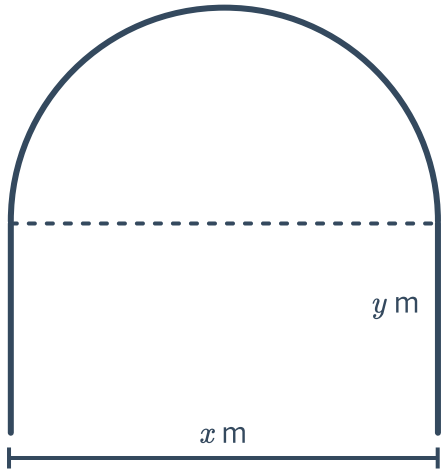
c Find $\frac{dA}{dx}$.

d Find the minimum possible area of the door.



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24 Pauline is building a plastic sheet green-house. The plan is to spread the plastic over 4 frames of heavy gauge wire. Each frame will have straight sides and a semicircular roof as shown in the diagram. She can only afford $64\ \text{m}$ of wire and wants to maximise the capacity of the green-house.



a Write an equation for y in terms of x.

b Find an expression for the cross-sectional area, A, of the frame in terms of x.

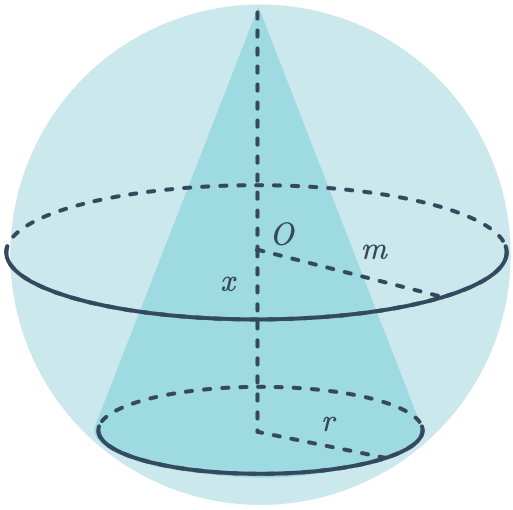
c Find $\dfrac{dA}{dx}$.

d For what value of x is $\dfrac{dA}{dx}$ equal to 0?

e If the minimum height of the straight sides is $1\ \text{m}$, find the maximum value for x.

f Find the maximum cross-sectional area of each frame, which would maximise the capacity of the green-house.

25 A cone is inscribed in a sphere of fixed radius m, centred at O. The height of the cone is x and the radius of the base is r, as shown in the diagram:



a Write an expression for the volume, V, of the cone in terms of m and x.

b Find the values of x for which the volume of the cone will be a maximum.

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26 The organisers of a fundraising event are trying to work out what they should charge per ticket to have the maximum number of people attend. They expect 1800 people if they charge $\$11$ per ticket. For each $\$0.25$ drop in price of the ticket, they expect an extra 180 people to attend.

a Write an expression for the price of a ticket after x drops in price of $\$0.25$ each drop.

b Write an expression for the number of people expected to attend after x drops of $\$0.25$ in the cost of a ticket.

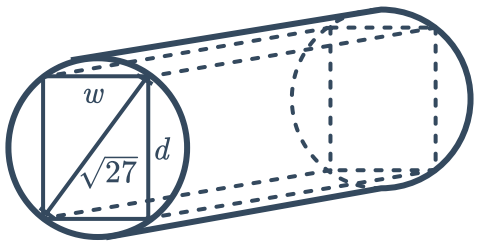
c Write an equation for the revenue, R, they can expect to receive in terms of x.

d Find the value of x that maximises the revenue.

e Find the price per ticket that will maximise revenue.

f Find the number of people that attend in order to maximise the revenue.

27 A rectangular beam of width w cm and depth d cm can be cut from a cylindrical log of wood as shown in the diagram below. The diameter of the cross-section of the log is $\sqrt{27}$ cm. The strength S of the beam is proportional to the product of its width and the square of its depth, so that $S = kd^2w$, where k is a positive constant.



a Show that $S = k\left(27w - w^3\right)$.

b Find the value of w that will give a beam of maximum strength.

c Find the value of d that will give a beam of maximum strength.

d Hence, find the maximum strength of a rectangular beam.

e A square beam with diagonal $\sqrt{27}$ cm is to be cut from an identical log. Find the strength of this beam.

f By what percentage is the maximum strength rectangular beam stronger than the square beam? Round your answer to one decimal place.

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